

Scalability & Future Plan

Date	3 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Single-user, sequential processing — no support for concurrent requests	Multiple simultaneous users would experience slower response times	Medium
2	Gemini API free tier limited to 5 requests/minute	AI feedback generation may be delayed or fall back to a generic message under heavy use	Medium
3	Free-tier hosting sleeps after 48 hours of inactivity	First request after inactivity has a longer cold-start delay	Low

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user, sequential pipeline on free CPU tier	Upgrade to paid Hugging Face hardware tier or add request queuing to support concurrent users
2	Data Storage	Reference concepts stored in a static JSON file	Migrate to a proper database (e.g., Firestore/PostgreSQL) to support dynamic concept addition and user history
3	Performance	~15-20 sec per analysis using CPU-only inference	Use GPU-backed hosting or smaller/quantized models to reduce inference time

4	Security	API key managed via environment variables/secrets	Add user authentication and rate-limiting at the application level for public multi-user deployment
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Add support for additional concepts and subjects	Short-term (1-2 months)	Broader applicability across more academic topics
Phase 2	Multi-language transcription and evaluation support	Medium-term (3-6 months)	Wider accessibility for non-English speakers
Phase 3	User accounts with progress tracking across multiple attempts	Medium-term (3-6 months)	Enables learners to monitor improvement over time
Phase 4	Integration with classroom/LMS platforms	Long-term (6-12 months)	Adoption by educators for structured classroom assessment